Introduction to Linked Lists

Q. What is a Linked List? Explain its representation in memory with the help of a suitable diagram and structure definition in C.

Definition to Remember

*A **Link*
is NOT
node, fo

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1. Structure of a Node

A standard linked list node has two components:

1. Data

Structure Definition in C:

struct N

2. Representation in Memory

  * The list is accessed via a special starting pointer called the **`head`** (or `start`).

  * If the

Diagram:

```
text
```

[Data|Next] [Data|Next] [Data|Next] NULL

(Node 1

3. Advantages over Arrays

  * **Dynamic Memory Allocation:** Size can grow and shrink at runtime; memory is allocated exactly as needed (no wastage).

  * **Eff

Must-Write Points (for 10 marks)

1. A Lin

Quick Recall

`Linked
points to